

Vaibhav Patil

DOB: 30/11/2003

B.Tech. in Mechanical Engineering

NIT Nagpur

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GitHub Profile

LinkedIn Profile

EDUCATION

• NIT Nagpur

B.Tech. in Mechanical Engineering

2021–25

CGPA: 8.45

EXPERIENCE

• MetLife | Software Engineer

Java, Spring Boot, React.js

Aug'25–Present

Hyderabad

- Modernized the legacy CDF platform from VB.NET monolith to Spring Boot microservices with React.js frontend
- Unified iLink, CDF, and SoldCase into a centralized GCA microservices platform
- Developed REST APIs with Spring Boot and DB2 integration for scalable backend communication
- Built responsive React.js frontend with optimized data fetching and UI performance
- Improved backend workflows and reduced customer onboarding time

• Turing | LLM Trainer

C++, Algorithms

Jul'25

Remote

- Designed Div1/Div2-level competitive programming problems to evaluate Alibaba Qwen's reasoning capabilities
- Developed end-to-end problem pipelines with constraints, C++ solutions, and test cases
- Performed adversarial testing on complex and edge-case scenarios to expose model failures
- Built automated test generators and analyzed model failures to improve LLM evaluation quality

• Zeus Numerix | SDE Intern

C++, Qt, QML

May'24–Jul'24

Pune

- Developed cross-platform GUI applications using Qt (QML) and C++
- Designed and deployed UI for PGK Simulation Software across Windows, Ubuntu, Android, and Raspberry Pi
- Implemented responsive UI components using Qt Creator and Qt Design Studio
- [GitHub](#), [Calculator Android APK](#) | [GitHub](#), [Desktop Deployed UI](#)

PROJECTS

• RAG-Based-Video-Learning-Assistant | Python, LLM, RAG

RAG-powered system for context-aware video search

Feb'26–Mar'26

- Built a RAG-based video learning assistant for searchable Q&A over video content
- Built a video-to-text pipeline using (FFmpeg → Whisper → chunking → JSON) processing
- Implemented bge-m3 embeddings and cosine similarity retrieval for relevant video search
- Integrated retrieved context into LLaMA prompts for context-aware response generation
- Improved response accuracy through chunk merging and LLM selection tuning
- [GitHub](#) | [Demo Video](#)

• AI-Powered News Research Assistant | Python, LangChain

AI-powered news research and semantic retrieval system

Jan'26–Feb'26

- Built an AI-powered news retrieval system for contextual article search.
- Developed a RAG pipeline using LangChain, OpenAI embeddings, and FAISS.
- Enabled low-latency, source-based semantic search and Q&A.
- [GitHub](#) | [Demo Video](#)

TECHNICAL SKILLS

AI/ML : Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG

Languages : Python, SQL, C++, Java, JavaScript

Libraries : NumPy, Pandas, Scikit-learn, LangChain

CS Fundamentals : DSA, OOPs, System Design, Computer Networks

Backend : Spring Boot, Node.js

Frontend : React.js

Database : MongoDB, MySQL

CERTIFICATIONS & ACHIEVEMENTS

- GitHub Copilot Certification (GH-300) - Microsoft | [Credential](#)
- The Ultimate Job Ready Data Science Course – CodeWithHarry | [Credential](#)
- Agile Delivery Model – Foundational Course | [Credential](#)
- Ranked 36 - [GFG](#) | Solved 400+ DSA problem - [Leetcode](#) | 2 Star rating - [Codechef](#)